

11.02.2025

Discussion On

- # Application of Database
- # Database building components
- # Data Abstraction

Database can provide best services to the users

Applications:

Ecommerce

Telecommunication

Airplane

Various Management Systems: Library, Bank, Hospital

Airport

Building components:

1. Character

↓

2. Field / attribute / Key / column

↓

3. Record / Entity

↓

4. Record set / Entity set

↓

5. Database

→ Data Abstraction

→ Database vs file system

16.02.25

Authorization - right to do something

Data abstraction - method of hiding information from unauthorized users
allowing someone to do something

3 level of hiding -

view level - user can see only permitted data

logical level -

physical level

view level user can use database in minimal way. Hiding/level is very high.
abstraction

logical level - administration level.

where there is no abstraction for data but the data structure is data fully abstracted

physical level - developer level. minimal data is needed.

Data independence -

logical level data independence
physical " " "

Making changes in logical level that doesn't affect physical level is called logical level data independence

23.02.25

File system Vs Database system

Problems of File system

1. Data redundancy and Inconsistency

2. Data isolation

3. Difficulty in accessing data

4. Concurrent Access Anomalies

5. Atomicity Problem

6. Security Problem

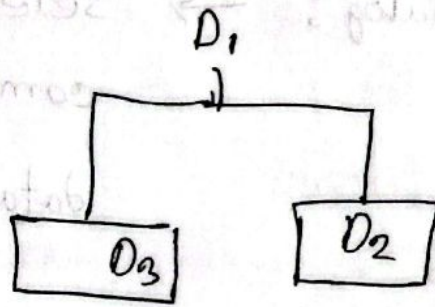
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Types of database

Database user

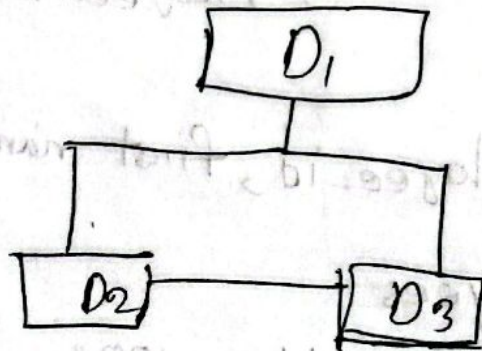
functions of DBA

Hierarchical database



IMS - Information Management System

Network database



There is internal connection between parents and children.

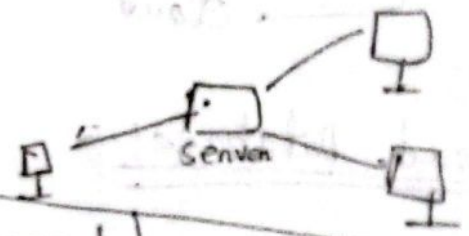
Relational database

organised in structured way

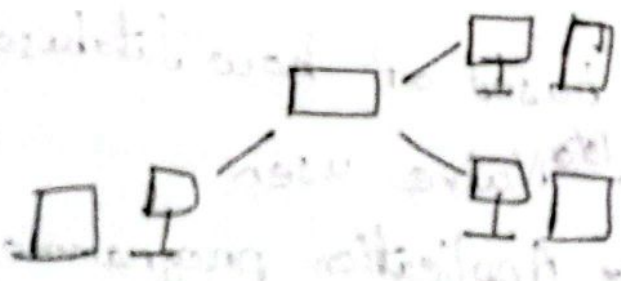
NO SQL: Non-relational database.

OO DB:
Object Oriented Database is, a database
where, ^{the} you have the ^{the} concept of OOP
like encapsulation, polymorphism

centralized Database:

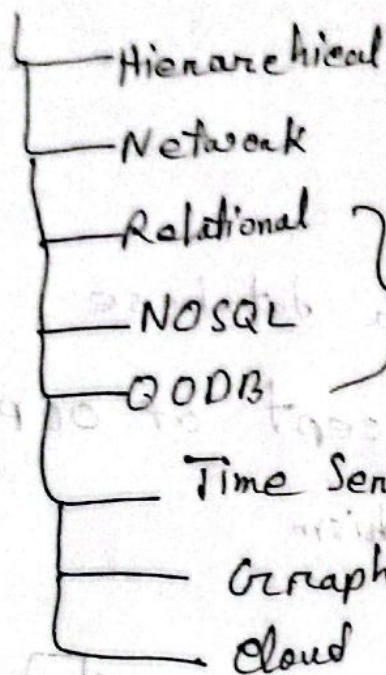


Distributed database: (widely used)



Time series database:

Graph database:



Centralised
Distributed

Types of Database

Cloud Database:

Database user: based of how database is used

- Naive user
- Application programmer
- Sophisticated
- Specialized

Naive user: end user

Application programmer:

RAD → Rapid Application Development

uses for testing purpose also

Sophisticated:

those who guides ~~time~~ in development
of some efficient system

Specializes: ~~user~~ Robot - user interface

0:04:03.25

→ Database Administrator (DBA)
→ Functional components of DB } - main book

Database Administrator - user

data & program to access those data

S/W → DBMS

⇓

DBA

Functions of DBA

Ø Schema Definition

Ø Access method definition

Ø Physical Organizational Modification

Ø Authorization

Ø Periodic maintenance - Backup

Up to date
Problem resolve

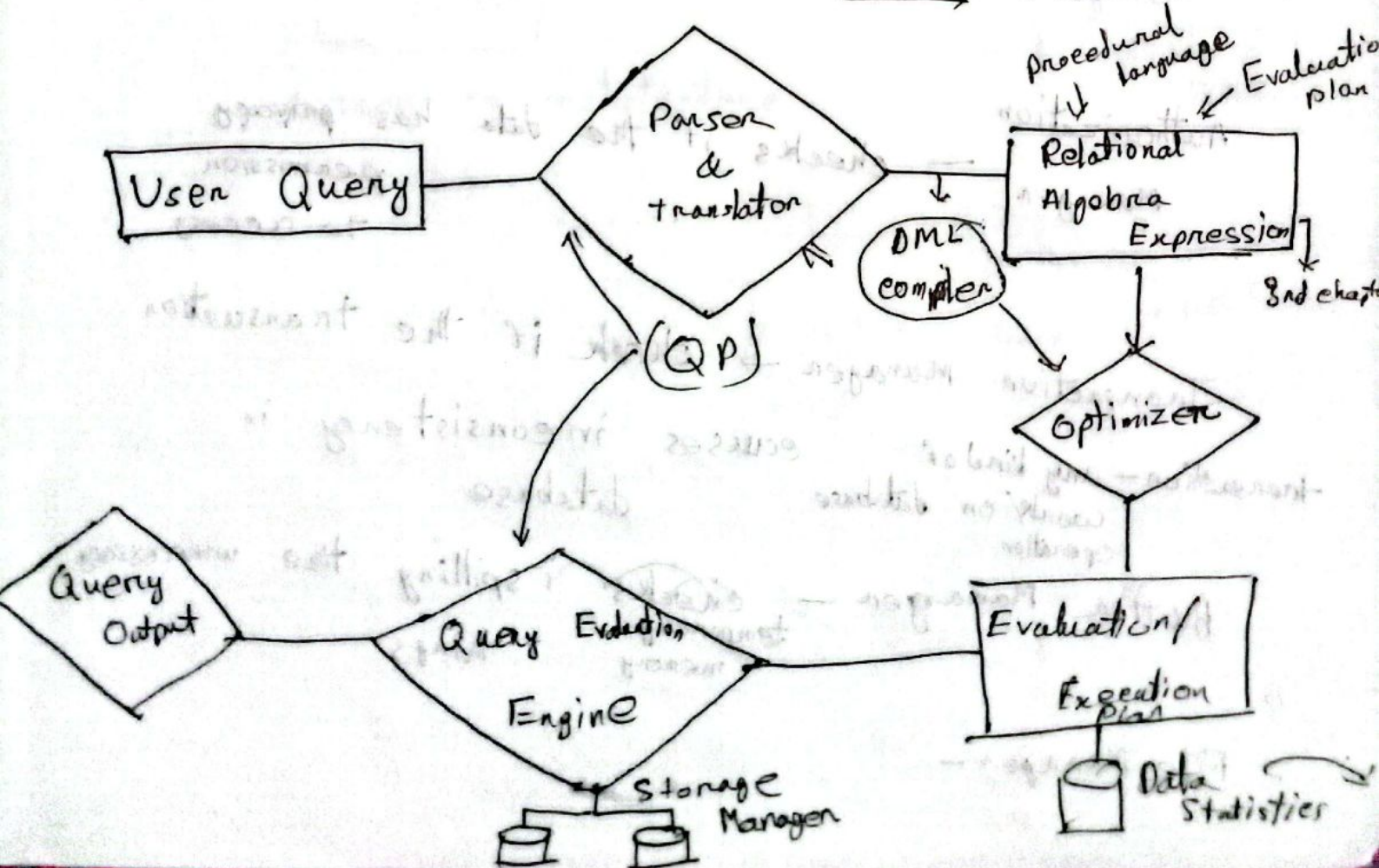
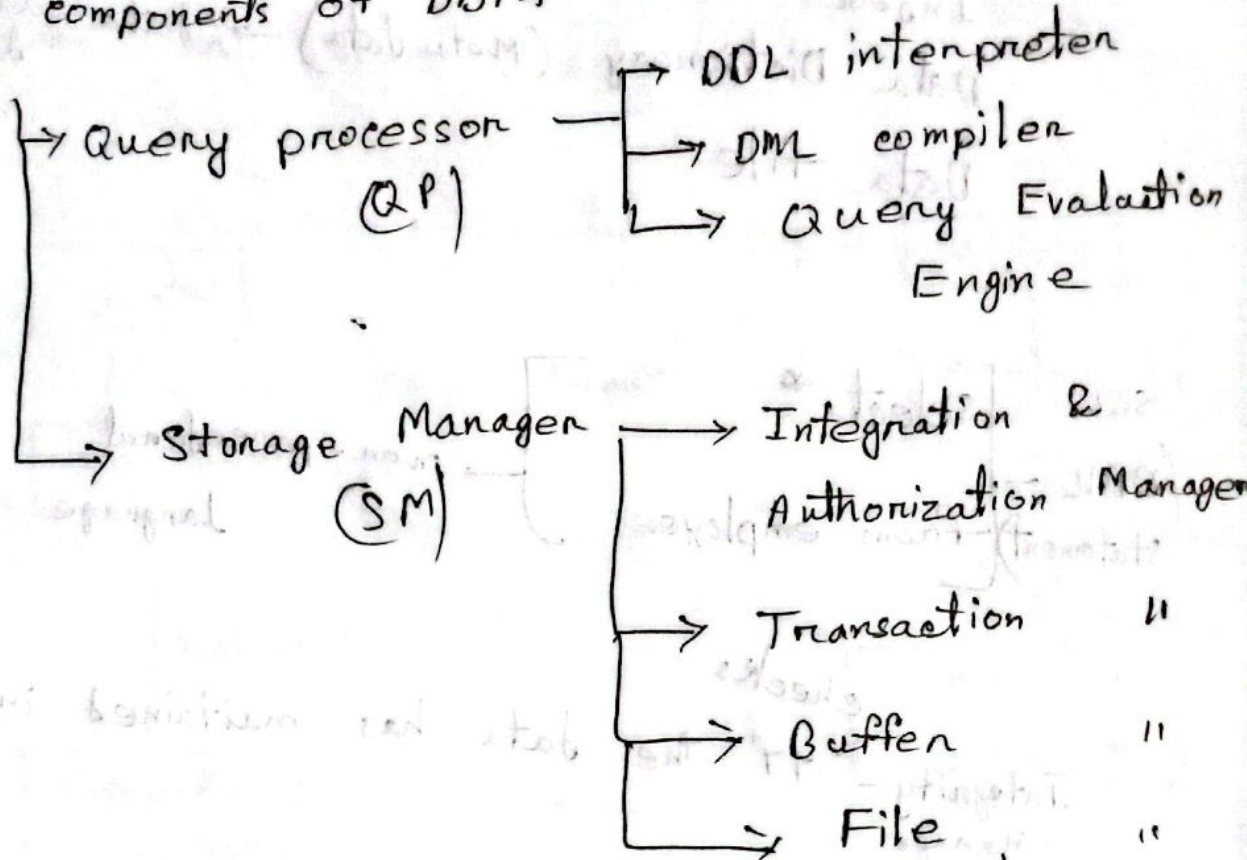
Schema - overall design of database

01.09.03.25

Building components of DBMS database:

Letter → character
word → Field/attribute/Key/column
sentence → Entity
paragraph → Entity set → Database
Document → Database

Functional components of DBMS



Data structures

Index

Data Dictionary (Meta data) → data about data

Data file

SQL
(DDL - statement) $\left[\begin{array}{l} \text{select *} \\ \text{from employees} \end{array} \right]$ non-procedural language

Integrity - checks
Manager - if the data has maintained integrity

Authorization
Manager

checks if the data has privacy permission to access

Transaction Manager → check if the transaction

transaction - any kind of work on database operation

causes inconsistency in database

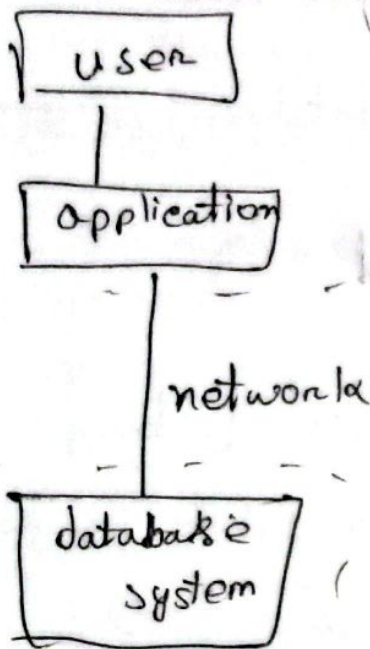
Buffer Manager - checks temporary memory if spilling the unnecessary things

File Manager -

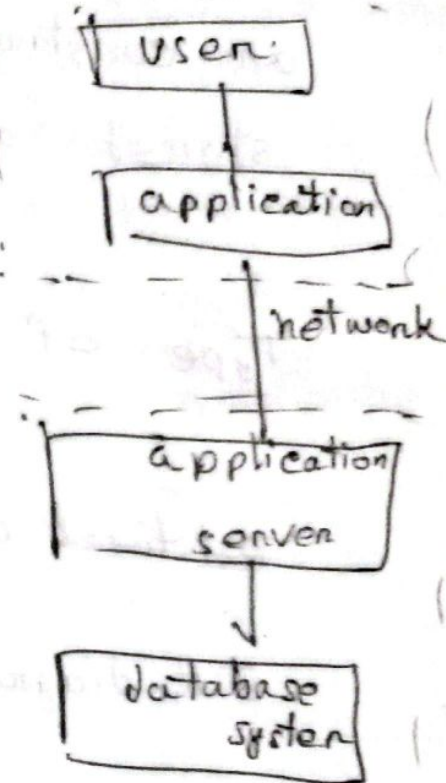
Architecture

(1) Two-tier

(2) Three-tier



a. two-tier architecture



b. three-tier architecture

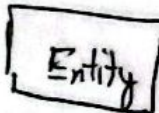
Database Model:

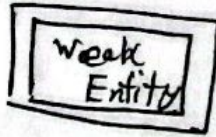
A database model shows the logical structure of a database, including the relationships and constraints that determine how data can be stored and accessed.

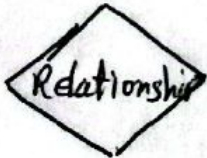
Type of database model

Relational database

E.R. diagram Symbols

Entity  → has primary own identity

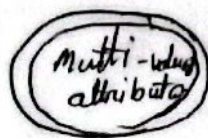
Weak Entity  → doesn't have own identity

Relationship  → relation between entities

Attribute 

Key attribute 

Key attribute



Derived attribute

attribute

(derived attribute)



derived from other attribute

———— Mandatory Relationship — Relationship is mandatory

- - - - Optional Relationship — is not mandatory relationship

—— partial Participation

===== total Participation

One to one



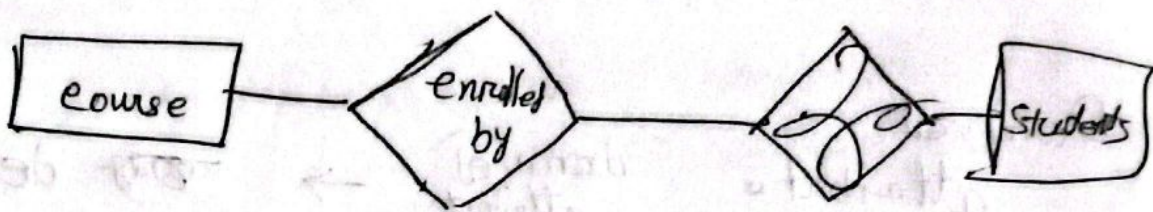
One to many



Many to one



Many to many



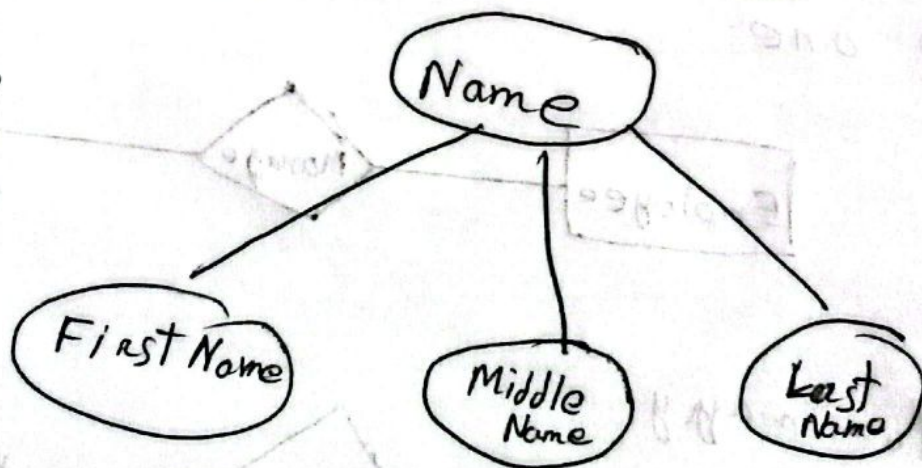
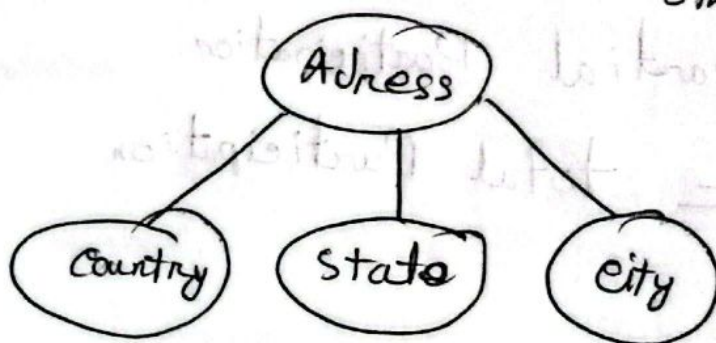
ଏକ attribute ଗଠନ compose ସଂଗଠିତ several other attribute ଗଠନ

ଏକ ଗଠନ composite attribute ଗଠନ

Example

A

composed from other simple attributes



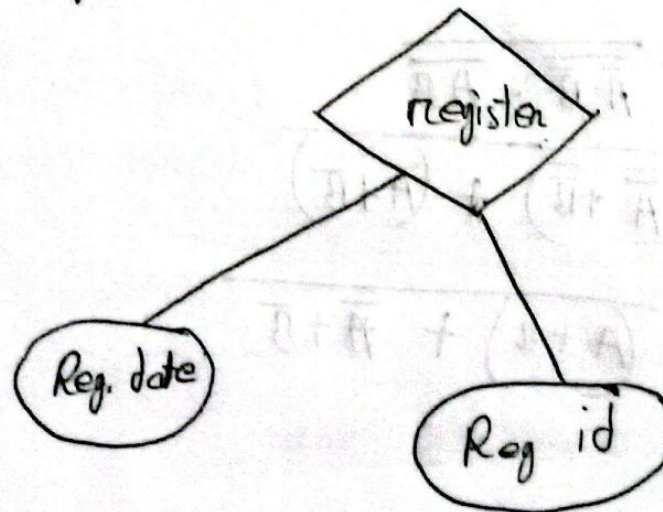
୦୦୦ → simple attribute

(Age) - derived attribute, depends on other attribute

Types of attribute

1. Simple vs composite
2. Single valued vs multi-valued
3. Stored vs derived
4. Key vs Non-Key attribute
5. Descriptive attribute

Example:



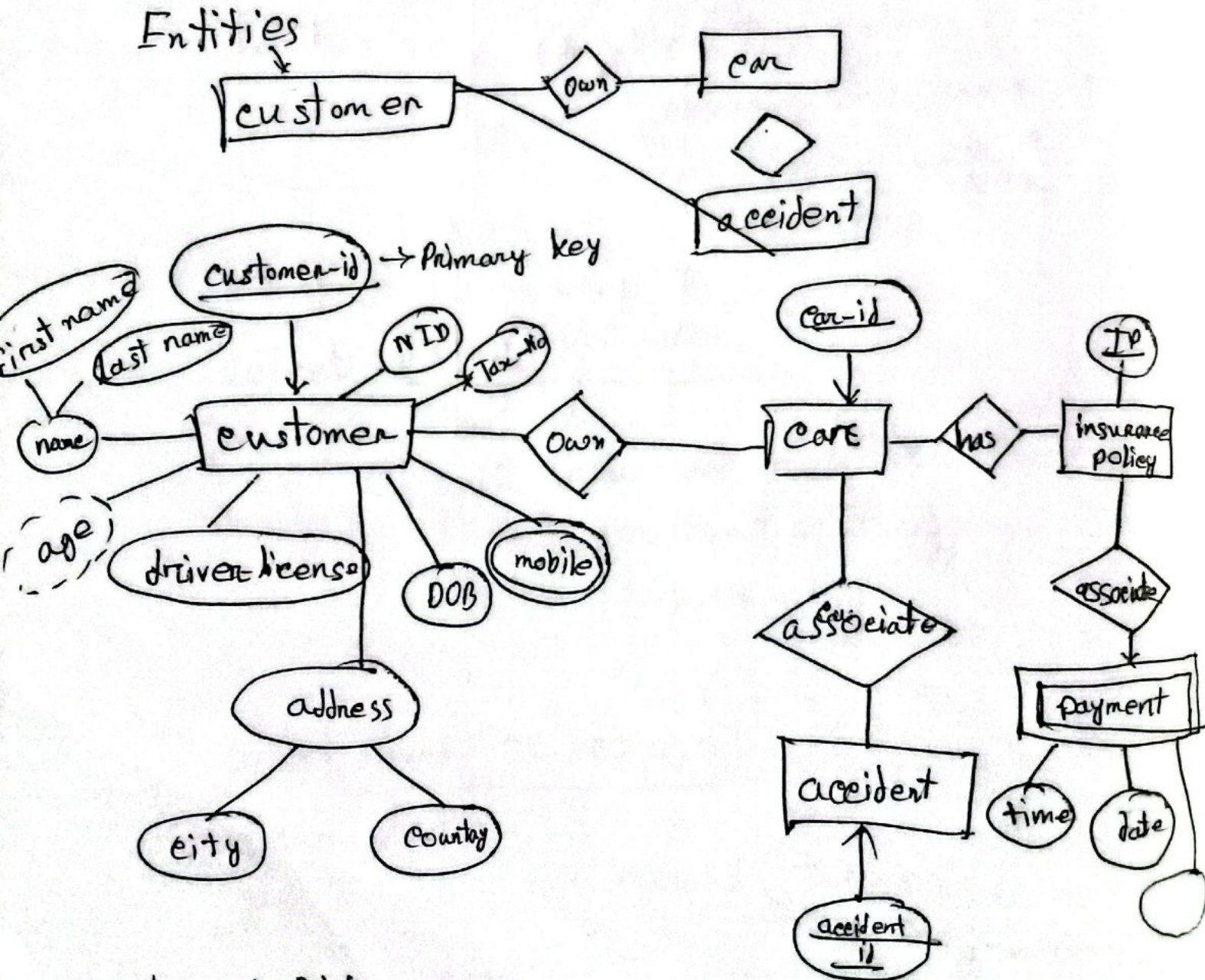
if relationship has attribute to describe it

D: 18.03.25

Create an ERD for car insurance Sub domain

company whose customer owns one or more cars with zero or more accidents. Each car has associated of recorded accident.

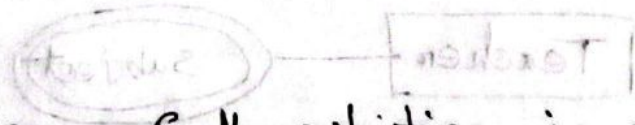
domain → entity
Relationship



DOB - Date of Birth

01. 21.09.25

Participation constraints



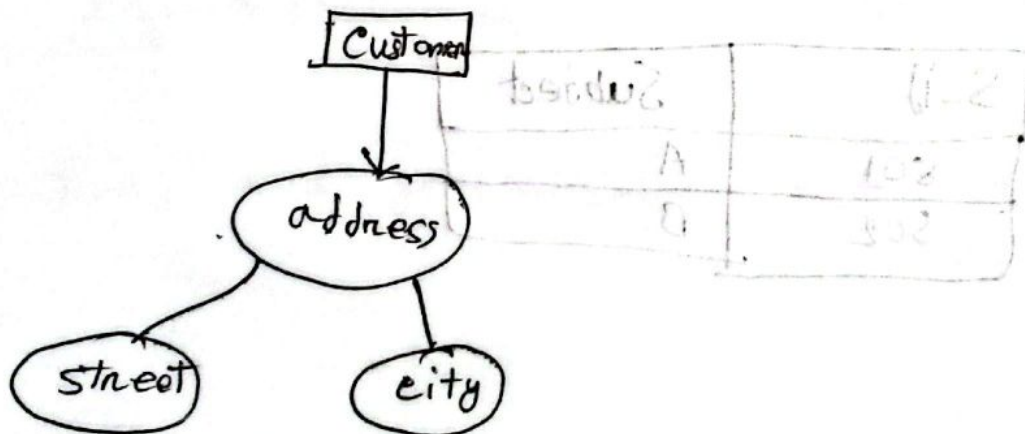
Total : Participation of all entities is mandatory

Partial : Participation of all entities not mandatory

Representation of ERD to table (Database)

Composite attribute

67	6-2
102	10
502	10



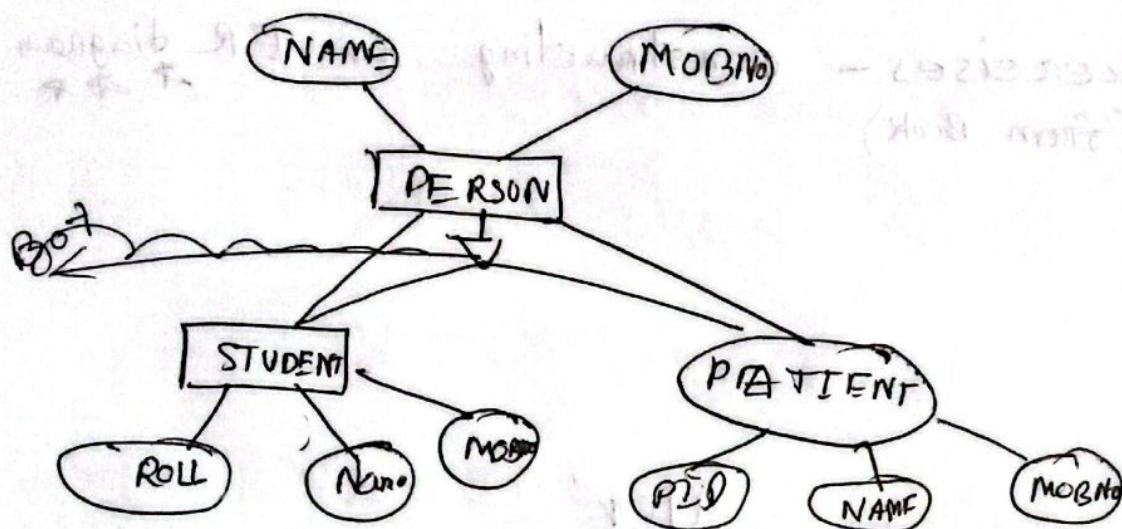
67	6-2
102	10
502	10

Address Table:

ID	Number	street	city
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Extended E-R Features:

Example of GENERALISATION



Specialization:

△ → Bottom-up

▽ → Top-down

specialization and generalisation - example from book (chapter-6)

Type of generalization -
attribute-defined
user defined

Total specialization or generalization
Partial specialization or generalization

Relational Algebra exercise konth

1. employee (person-name, street, city)
works (person-name, company name, salary)

Find names of all the employees
on city 'Miami'

SQL

select name
from employees

where city = 'Miami'

$\pi_{\text{name}} (\sigma_{\text{city} = \text{'Miami'}} (\text{employees}))$

Find the salary and city of all employees

select ^{works.}person-name, salary, city
from employee, works
where works.person-name =
employee.person-name

$\pi_{\text{person_name}, \text{salary}, \text{city}} (\sigma_{\text{works. person_name} = \text{employee. person_name}}$

$(\text{employee} \times \text{works}))$